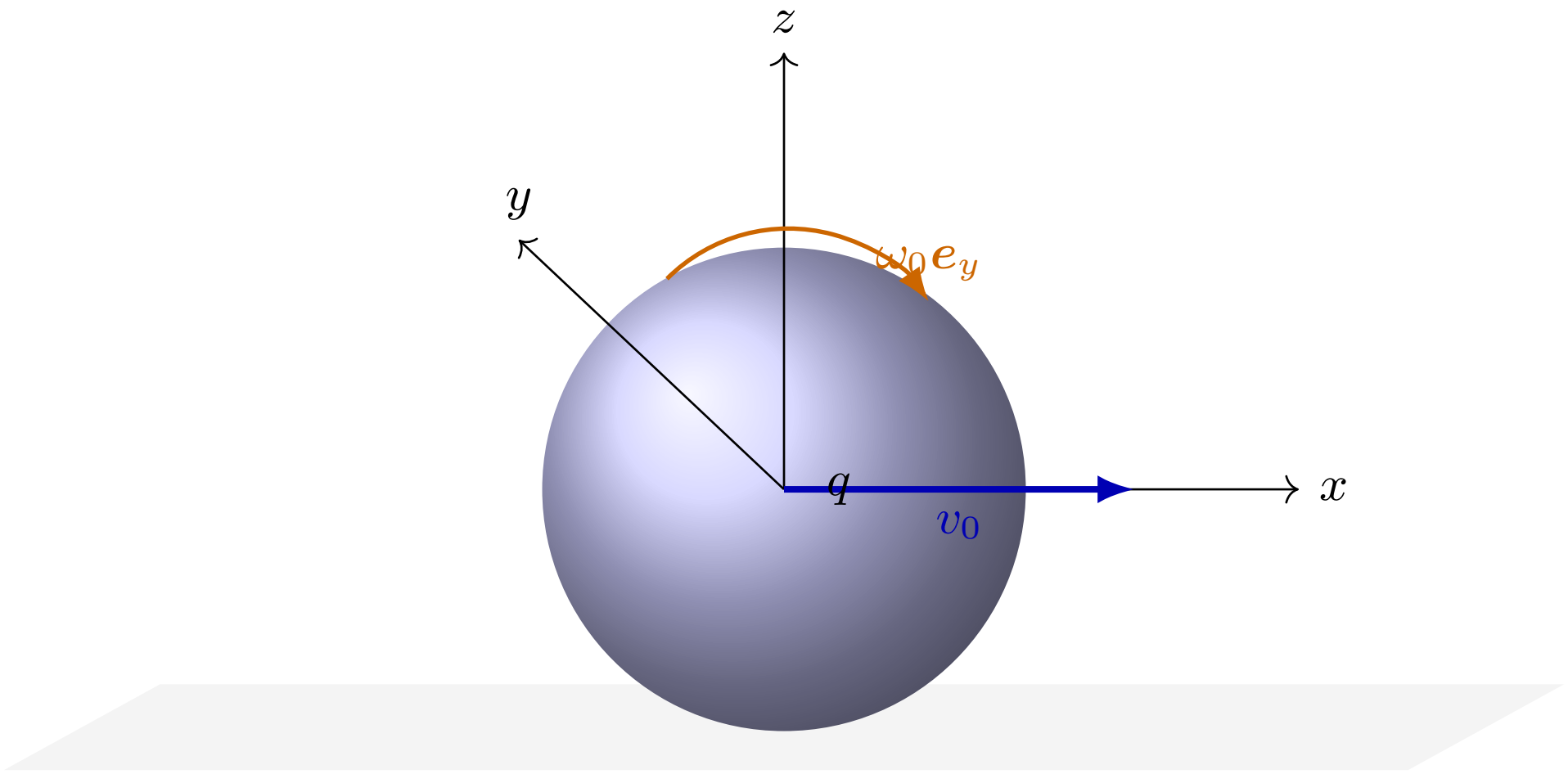

$$\boldsymbol{B} = -B\boldsymbol{e}_z$$